## 1. Preparations

• Give the definition of Newton polygon for a power series for a formal power series in one variable, with values in a non-archimedean field. (Use AI to generate needed background knowledge.)

• Give lemma to show that Newton polygon of the product of two power series is the Minkowski sum of the Newton polygons of the two power series.

**Proposition 1.1.** *Let $F(t) = \sum_{n\geq 0} f_n t^n$ and $G(t) = \sum_{n\geq 0} g_n t^n$ be two power series in $\mathbb{Q}_p[\![t]\!]$. Then the Newton polygon of $F(t)G(t)$ is the Minkowski sum of the Newton polygons of $F(t)$ and $G(t)$.*

## 2. Formulation of the local ghost series

• Input data: $a \in \{1, \dots, p-4\}$. Fixed throuhgout.

• Notation from the paper $\Delta = \mathbb{F}_p^\times$. Also we need the Teichmüller character $\omega$ of $\mathbb{F}_p^\times$ to $\mathbb{Z}_p^\times$.

• We have an equivalent pair of definitions:

(1) $\varepsilon = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon}$ of $\Delta^2$ (relevant to $\bar{\rho}$). (no need to explain what $\bar{\rho}$ is.)

(2) A variable $s_\varepsilon \in \mathbb{Z}/(p-1)\mathbb{Z}$, usually represented by an element in $\{0, 1, \dots, p-2\}$.

• Define $k_\varepsilon \in \{2, \dots, p\}$ to be an integer such that $k_\varepsilon \equiv a + 2s_\varepsilon + 2 \bmod p-1$. (See notation 2.24)

• (Proposition 4.1) Turn the main result of this proposition into a definition of these dimensions. (DO NOT PROVE IT.)

Note also the special case $k = k_\varepsilon + k_\bullet(p-1)$ when $k \equiv k_\varepsilon \bmod (p-1)$. there is an abbreviation $d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1)$ for $d_k^{\mathrm{Iw}}(\varepsilon \cdot (1 \times \omega^{2-k}))$ (where $\varepsilon_1$ means the first factor of $\varepsilon$).

• defined $\delta_\varepsilon$ is Notation 4.2.

• Prove Corollary 4.4 (which is trivial).

• In §4.6, ignore (4.6.1), but define $t_1^{(\varepsilon)}$ and $t_2^{(\varepsilon)}$ as in §4.6, yet note that $t_i$ depends on $a$ (so need to be careful when writing LEAN.)

• Proposition 4.7 is taken as DEFINITION. Do not prove it.

• Now, turn to Definition 2.25 of ghost series. The subscript $\widetilde{\mathrm{H}}$ can be ignored, or just replaced by $(a)$. Here $\varepsilon$ means $\varepsilon_1 \times \varepsilon_2 = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon}$ as above.

Also verify (2.25.3) and (2.25.4).

• Need Notation 4.10.

• We need some definition of degree of basis elements (without knowing what they are, just taken as symbols), this is in Notation 3.8(2)(3).

• For Proposition 4.11, this is a medium size proof (a good size for AI to handle). (I am pretty sure it is correct; no need to verify its correctness, but maybe a good exercise for AI.)

• Please also formalize the statement in Remark 4.13.

• No need to formalize Corollary 4.14.

• Give Notation 4.17. (Note that this is the same as removing the factors of ghost series at the weights in $k$.

• Proposition 4.18 is once again a medium size proof (a good size for AI to handle). (I am pretty sure it is correct; no need to verify its correctness, but maybe a good exercise for AI.)

• Define Notation 4.27, and prove Proposition 4.28.

## 3. More difficult results on ghost series

Formalize everything in section 5.